R Notebook

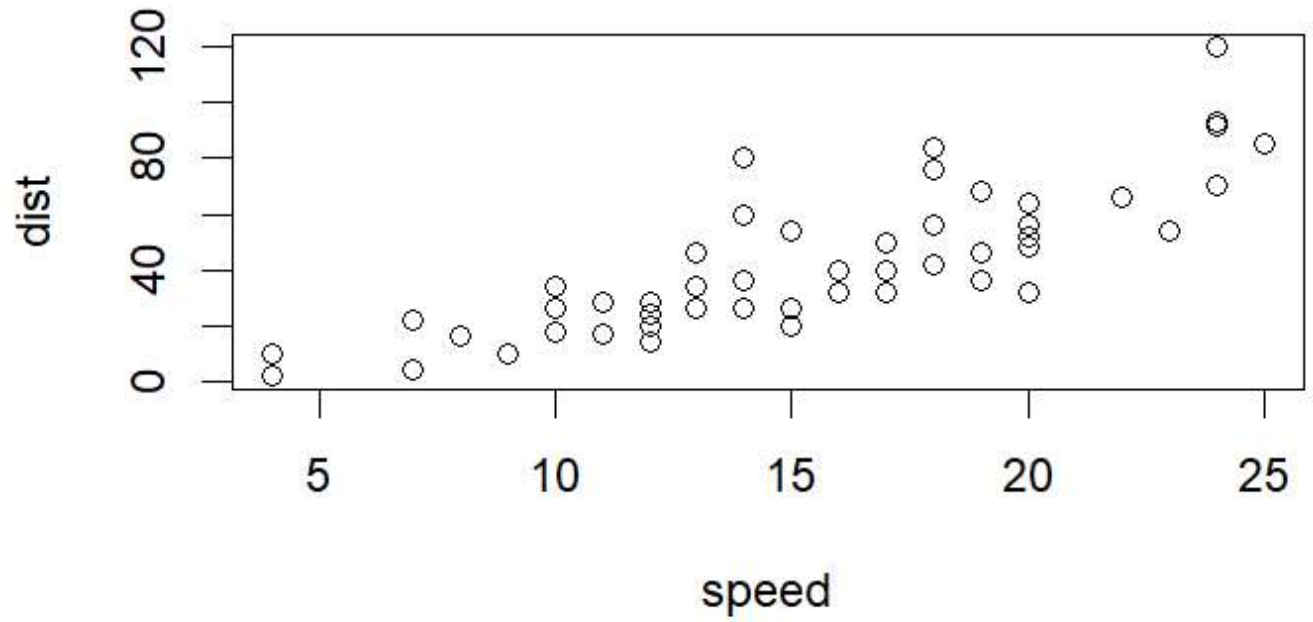
Code

This is an R Markdown (http://rmarkdown.rstudio.com) Notebook. When you execute code within the notebook, the results appear beneath the code.

Try executing this chunk by clicking the *Run* button within the chunk or by placing your cursor inside it and pressing *Ctrl+Shift+Enter*.

Hide

```
plot(cars)
```



Add a new chunk by clicking the *Insert Chunk* button on the toolbar or by pressing *Ctrl+Alt+I*.

When you save the notebook, an HTML file containing the code and output will be saved alongside it (click the *Preview* button or press *Ctrl+Shift+K* to preview the HTML file).

The preview shows you a rendered HTML copy of the contents of the editor. Consequently, unlike *Knit*, *Preview* does not run any R code chunks. Instead, the output of the chunk when it was last run in the editor is displayed.

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```
library(tidyverse)
setwd("C:\\Users\\adity\\Downloads\\P_Data_Extract_From_World_Development_Indicators")
```

```
Warning: The working directory was changed to C:/Users/adity/Downloads/P_Data_Extract_From_World
_Development_Indicators inside a notebook chunk. The working directory will be reset when the ch
unk is finished running. Use the knitr root.dir option in the setup chunk to change the working
directory for notebook chunks.
```

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```
list.files()
```

```
[1] "a7c69636-11fe-496e-a19a-fa3dbe3cd40f_Series - Metadata.csv"
[2] "female.csv"
```

Hide

```
female <- read.csv(file = "female.csv")
female <- female %>%
  filter(Country.Name == "Brazil" | Country.Name == "India" | Country.Name == "United States")

rel_col <- which(colnames(female)=="Country.Name" | colnames(female)== "X2014..YR2014." | colnam
es(female)== "X2017..YR2017." | colnames(female) == "X2021..YR2021." )
rel_col
```

```
[1]  3  7 10 14
```

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```
female <- female[rel_col] %>%
  rename("2014" = 2, "2017" = 3, "2021" = 4)

transpose_f <- data.frame(t(female[-1]))
colnames(transpose_f) <- female[, 1]

transpose_f$Brazil = as.numeric(as.character(transpose_f$Brazil))
transpose_f$India = as.numeric(as.character(transpose_f$India))
transpose_f$"United States" = as.numeric(as.character((transpose_f$"United States")))

female <- female %>% distinct(Country.Name, .keep_all = TRUE)


head(transpose_f)
```

	Brazil <dbl>	India <dbl>	United States <dbl>
2014	64.77	43.13	94.80
2017	67.51	76.64	92.69
2021	80.87	77.55	96.79

3 rows

Hide

```
setwd("C:\\Users\\adity\\Downloads\\P_Data_Extract_From_World_Development_Indicators (1)")
```

```
Warning: The working directory was changed to C:/Users/adity/Downloads/P_Data_Extract_From_World
_Development_Indicators (1) inside a notebook chunk. The working directory will be reset when th
e chunk is finished running. Use the knitr root.dir option in the setup chunk to change the work
ing directory for notebook chunks.
```

Hide

```
list.files()
```

```
[1] "704a7eca-5465-4ed1-88f6-e74a504e8347_Series - Metadata.csv"
[2] "male.csv"
```

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```
male <- read.csv(file = "male.csv")
male <- male %>%
  filter(Country.Name == "Brazil" | Country.Name == "India" | Country.Name == "United States")

male <- male[rel_col] %>%
  rename("2014" = 2, "2017" = 3, "2021" = 4)

transpose_m <- data.frame(t(male[-1]))
colnames(transpose_m) <- male[, 1]

transpose_m$Brazil = as.numeric(as.character(transpose_m$Brazil))
transpose_m$India = as.numeric(as.character(transpose_m$India))
transpose_m$"United States" = as.numeric(as.character((transpose_m$"United States")))

male <- male %>% distinct(Country.Name, .keep_all = TRUE)

head(transpose_m)
```

	Brazil <dbl>	India <dbl>	United States <dbl>
2014	71.69	62.76	92.36
2017	72.85	83.01	93.57
2021	87.05	77.51	93.09

3 rows

Hide

```
transpose_m <- rename(transpose_m, "Brazil_m" = 1, "India_m" = 2, "United_States_m" = 3)

transpose_m <- rownames_to_column(transpose_m, var="Year")
transpose_f <- rownames_to_column(transpose_f, var="Year")

acct_owner_by_gender <- merge(x = transpose_m, y = transpose_f, by = "Year", all.x = TRUE)
acct_owner_by_gender <- rename(acct_owner_by_gender, "United_States" = 7)
head(acct_owner_by_gender)
```

	Year <chr>	Brazil_m <dbl>	India_m <dbl>	United_States_m <dbl>	Brazil <dbl>	India <dbl>	United_States <dbl>
1	2014	71.69	62.76	92.36	64.77	43.13	94.80
2	2017	72.85	83.01	93.57	67.51	76.64	92.69
3	2021	87.05	77.51	93.09	80.87	77.55	96.79

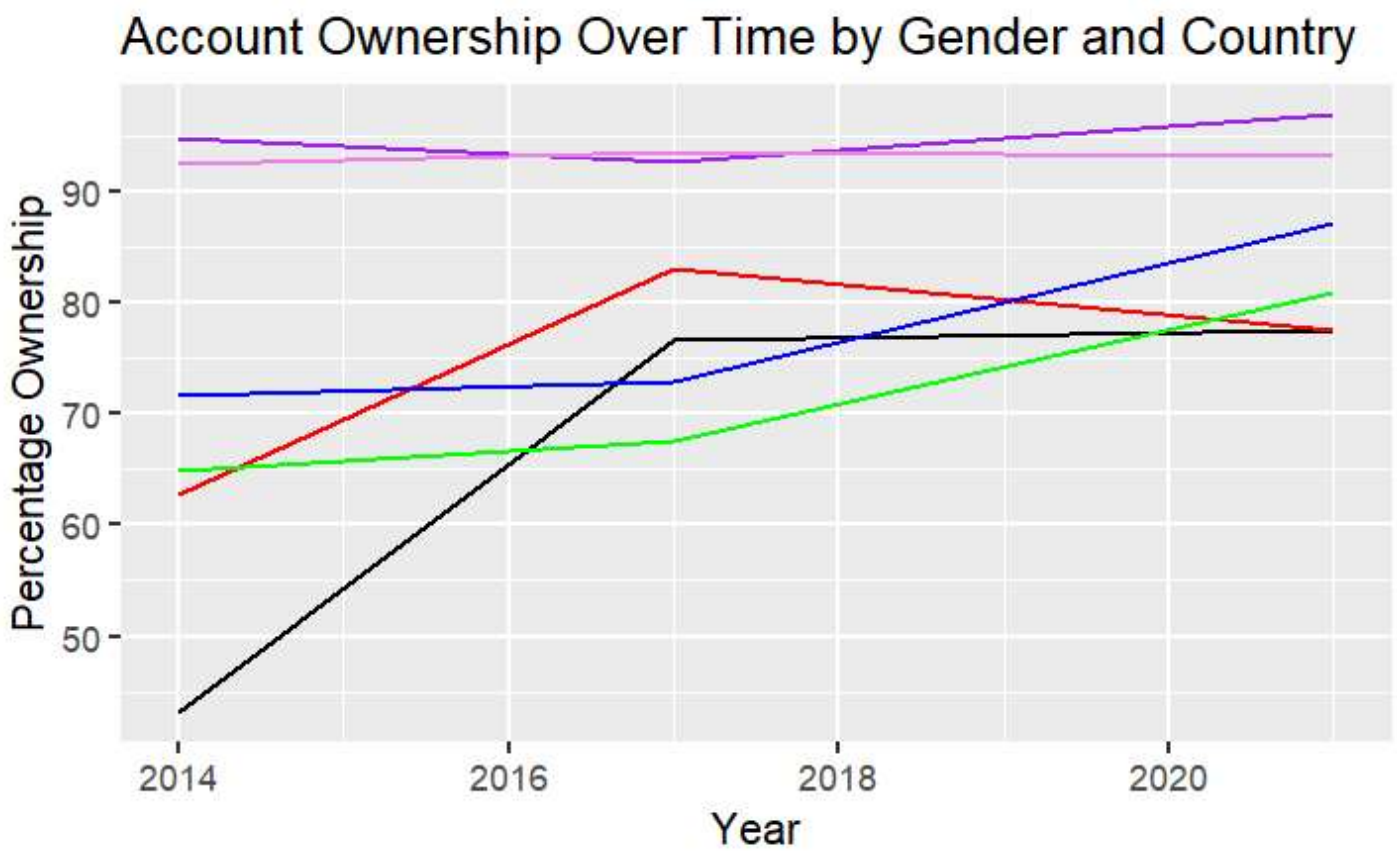
3 rows

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```
acct_owner_by_gender$Year <- as.numeric(acct_owner_by_gender$Year)

gfg_plot <- ggplot(acct_owner_by_gender, aes(x = Year)) +
    geom_line(aes(y = India), color = "black") +
    geom_line(aes(y = India_m), color = "red") +
    geom_line(aes(y = Brazil), color = "green") +
    geom_line(aes(y = Brazil_m), color = "blue") +
    geom_line(aes(y = United_States), color = "purple") +
    geom_line(aes(y = United_States_m), color = "violet") +
    labs(x = "Year", y = "Percentage Ownership",
       title = "Account Ownership Over Time by Gender and Country")

gfg_plot
```



2. Drawing Conclusions: United States: Both male and female ownership rates are consistently above 90%. There is only a small gap between male and female percentages, indicating relatively minimal gender disparity.

Brazil: Both genders show a steady increase from 2014 to 2021. Although men consistently have slightly higher ownership rates, the gap remains modest (roughly 5–7 percent).

India: Exhibits the largest jump between 2014 and 2017 (for both men and women). After 2017, male ownership shows a slight dip by 2021, while female ownership continues a modest rise—leading to a narrower gender gap by 2021 compared to earlier years.

3.Cherry Picking: - If you remove certain years, the conclusion about a large jump may disappear. Suggesting a more modest rise overall. - In Brazil, ignoring 2017 might not dramatically change the overall rising trend. - In the United States, removing any single year barely affects the conclusion, since the ownership rates are already high and stable.

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```
library(tidyverse)

setwd("C:\\Users\\adity\\Downloads\\P_Data_Extract_From_World_Development_Indicators (2)")
```

```
Warning: The working directory was changed to C:/Users/adity/Downloads/P_Data_Extract_From_World
_Development_Indicators (2) inside a notebook chunk. The working directory will be reset when th
e chunk is finished running. Use the knitr root.dir option in the setup chunk to change the work
ing directory for notebook chunks.
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```
primary <- read.csv(file = "primary.csv")

setwd("C:\\Users\\adity\\Downloads\\P_Data_Extract_From_World_Development_Indicators (3)")
secondary <- read.csv(file = "secondary.csv")

primary <- primary[primary$Country.Name %in% c("Canada","Germany","France","Japan","South Kore
a"), ]
secondary <- secondary[secondary$Country.Name %in% c("Canada","Germany","France","Japan","South
Korea"), ]

rel_col <- which(colnames(primary) %in% c("Country.Name","X2014..YR2014.","X2017..YR2017.","X202
1..YR2021."))

primary <- primary[, rel_col]
colnames(primary)[2:4] <- c("2014","2017","2021")

secondary <- secondary[, rel_col]
colnames(secondary)[2:4] <- c("2014","2017","2021")
```

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```
transpose_primary <- data.frame(t(primary[,-1]), stringsAsFactors = FALSE)
colnames(transpose_primary) <- primary$Country.Name

transpose_secondary <- data.frame(t(secondary[,-1]), stringsAsFactors = FALSE)
colnames(transpose_secondary) <- secondary$Country.Name

for(country in colnames(transpose_primary)) {
  transpose_primary[[country]] <- as.numeric(transpose_primary[[country]])
}
for(country in colnames(transpose_secondary)) {
  transpose_secondary[[country]] <- as.numeric(transpose_secondary[[country]])
}

transpose_primary$Year <- rownames(transpose_primary)
transpose_secondary$Year <- rownames(transpose_secondary)

transpose_primary$Year <- as.numeric(transpose_primary$Year)
transpose_secondary$Year <- as.numeric(transpose_secondary$Year)

head(transpose_primary)
```

	Canada <dbl>	France <dbl>	Germany <dbl>	Japan <dbl>	Year <dbl>
2014	100.00	83.36	97.64	83.66	2014
2017	94.73	79.82	100.00	91.43	2017
2021	90.16	95.87	100.00	94.10	2021

3 rows

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```
head(transpose_secondary)
```

	Canada <dbl>	France <dbl>	Germany <dbl>	Japan <dbl>	Year <dbl>
2014	99.57	97.26	98.80	99.67	2014
2017	99.87	97.23	99.30	99.49	2017
2021	99.88	99.76	99.97	99.22	2021

3 rows

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```
transpose_primary <- transpose_primary[order(transpose_primary$Year), ]

x_vals <- transpose_primary$Year

plot(x_vals, transpose_primary$Canada,
     type="l", lwd=2, col="red", ylim=c(50,100),
     xlab="Year", ylab="Percentage Ownership",
     main="Account Ownership: Primary Education or Less")

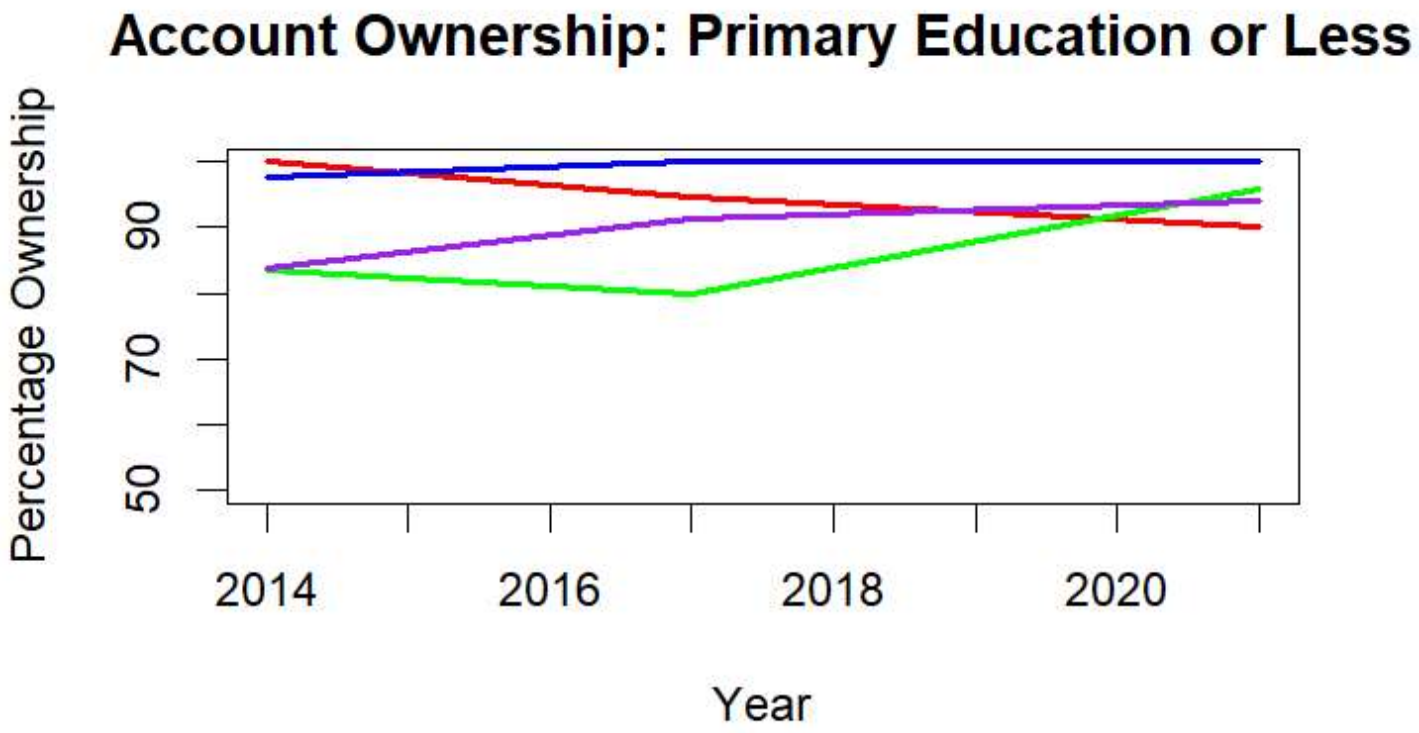
lines(x_vals, transpose_primary$Germany, col="blue", lwd=2)
```

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```
lines(x_vals, transpose_primary$France,  col="green",lwd=2)
lines(x_vals, transpose_primary$Japan,   col="purple",lwd=2)
```

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```
lines(x_vals, transpose_primary$`South Korea`, col="orange",lwd=2)
```



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```
transpose_secondary <- transpose_secondary[order(transpose_secondary$Year), ]
x_vals2 <- transpose_secondary$Year

plot(x_vals2, transpose_secondary$Canada,
     type="l", lwd=2, col="red", ylim=c(90,100),
     xlab="Year", ylab="Percentage Ownership",
     main="Account Ownership: Secondary Education or More")

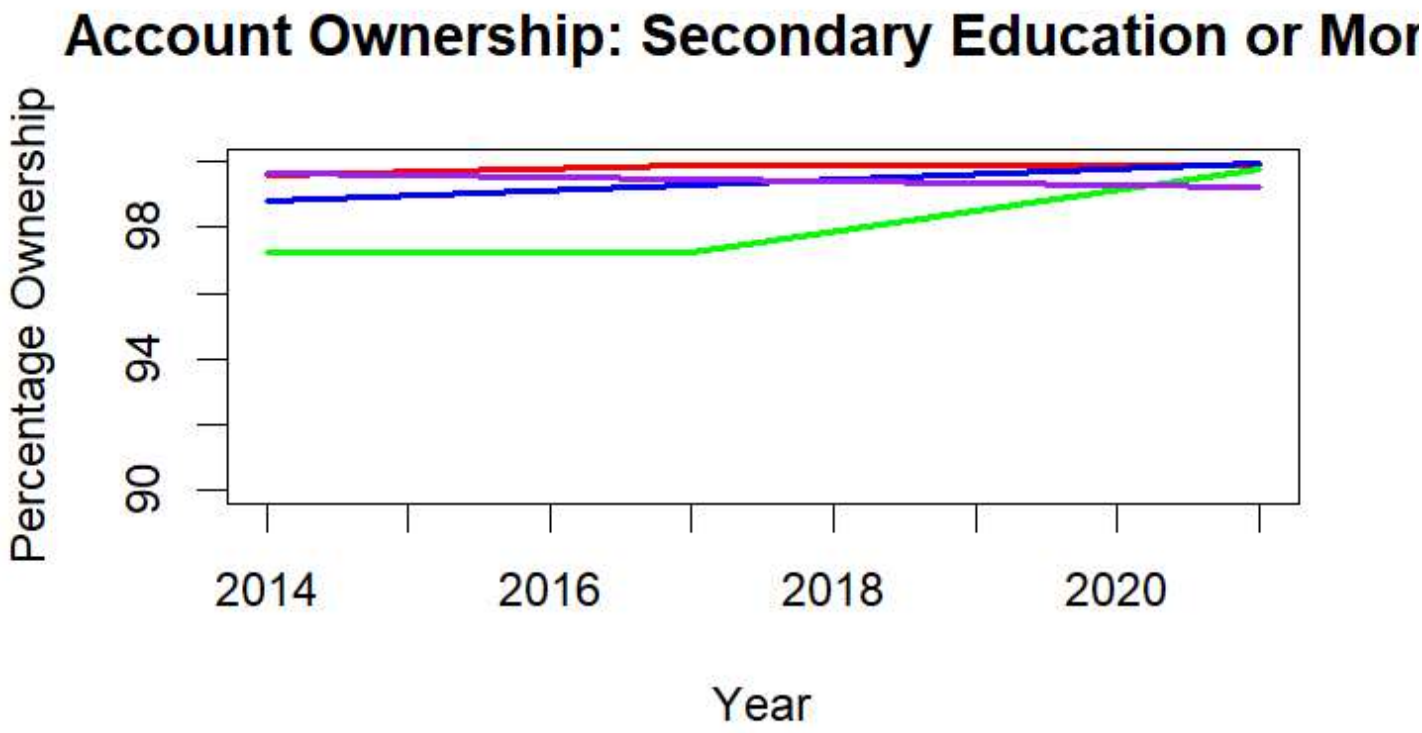
lines(x_vals2, transpose_secondary$Germany, col="blue", lwd=2)
```

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```
lines(x_vals2, transpose_secondary$France,  col="green",lwd=2)
lines(x_vals2, transpose_secondary$Japan,   col="purple",lwd=2)
```

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```
lines(x_vals2, transpose_secondary$`South Korea`, col="orange",lwd=2)
```



Summary (Primary Education): In our five choosen countries (Canada, Germany, France, Japan, and South Korea), the data shows that people with primary education or less have a moderate number of bank accounts. Most countries show an incresing trend from 2014 to 2021, even though the exact numbers differ. Japan and Germany sometimes have higher percentages than South Korea or France, which may mean they have better acces to banks for people with less education.

Summary (Secondary Education): For people with secondary education or more, the numbers are higher in all five countries. This shows that more education helps in getting bank accounts. The gap between 2014 and 2021 is also larger in some countries, which may be due to econmic changes or new policies.

Overall Conclusion: Education level seems to have a big impact on financial inclusion. In developed countries, people with more education are more likely to have bank accounts. Both groups show an upward trend, but the ones with higher education always have more bank accounts. This folow shows that education helps people learn about and use banking services.